

FORS351 - Lab 1: Exploring Spatial, Spectral, and Temporal Resolution

Name: Theo Symonds

Platforms Used:

1. Copernicus Browser (Sentinel-2): <https://browser.dataspace.copernicus.eu/>
2. ESRI Living Atlas – Landsat Explorer: <https://livingatlas.arcgis.com/landsatexplorer/>

Part 1 – Choose a Study Location:

1. Location name: Nogent-le-Rotrou, France
2. Approximate latitude/longitude: 48.3225, 0.8225 (WGS84)
3. Why you chose it (1–2 sentences): I chose this location because it is my hometown, and so I have some context of the area which I think will help me better understand the different things we are looking at throughout this lab. I'm also excited to see a place I know so well from the ground, from above, and I think I will see a few things differently than I had before.

Part 2 – Sentinel-2 (Copernicus Browser)

4. Paste Screenshot of summer image here (2 pts):



5. Paste screenshot of winter image here (2 pts):



Answer:

6. Which bands make up true color (2 pts)?

Red, Green, and Blue

7. What features are easiest to see (2 pts)?

Due to the rural setting is easiest to see the fields surrounding the town, and it is also easy to see the roads and building placement within the town.

Normalized Difference Vegetation Index:

8. Paste Screenshot of summer image here (2 pts):



9. Paste screenshot of winter image here (2 pts):



Answer:

10. What does NDVI represent (2 pts)?

Normalized Difference Vegetation Index (NDVI) represents the degree of 'greenness' of vegetation, which is used as an indicator for vegetation health.

11. How does NDVI change seasonally (2 pts)?

For the bushes and forested vegetation (like the hedgerows between fields and small pockets of forest), the NDVI increases in spring and peaks in summer, and tapers off in fall and is weakest in winter. However we see something slightly different for the agricultural fields which are mostly wheat, where these fields are actually 'greener' in the winter than they are in the summer when the yellowish/brown wheat is at its peak growing season.

Temporal resolution:

Scroll through dates

12. How often is imagery available at your location (2 pts)?

The imagery seems to be available at least 1-2 times a week for cloud cover above 80%, because the region is frequently cloudy. If we want less than 80% cloud cover, the temporal resolution has a sharp drop off. Even reducing cloud cover to 75% and below made the temporal resolution less than once a week, and anything below 25% cloud cover was only available once or twice a month, and some months didn't have any day with less than 25% cloud cover. There was more cloud cover in winter than summer.

13. How does cloud cover affect this (2 pts)?

Cloud cover will reduce the amount of imagery that can be used and compared across time. If you wanted to monitor a very specific spot (ex: a building) in imagery every week for a year, the

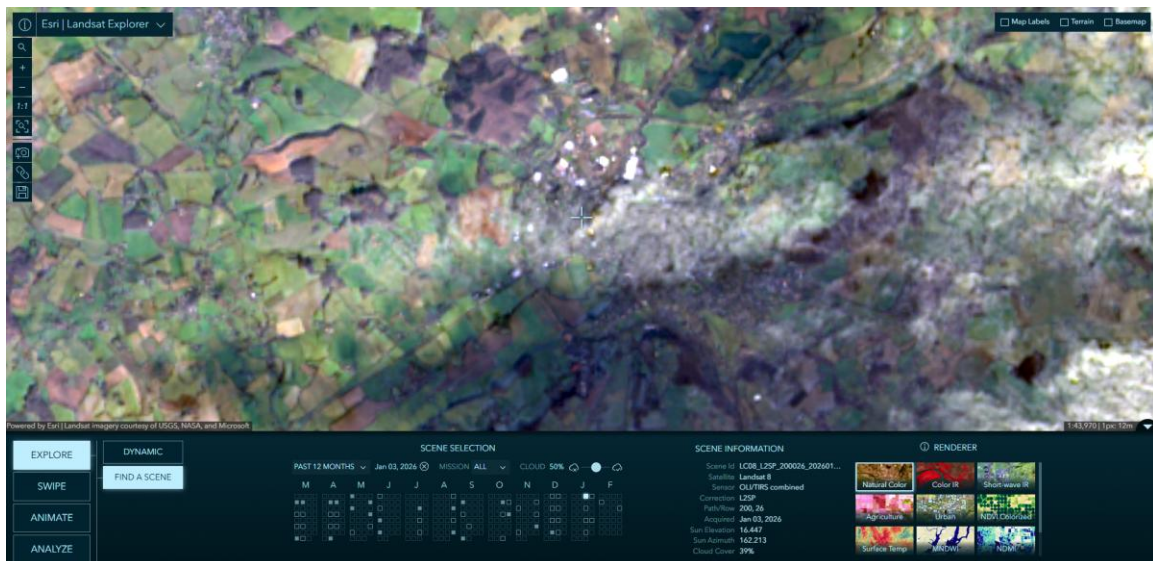
biggest impediment would be that even small amount of cloud cover could obfuscate the building you want to look at on the days the satellite image is taken, and so the amount of usable images for each week over the year gets exponentially reduced due to the percentage cloud cover (fewer and fewer available images with every desired decreased of cloud cover percentage).

Part 3 – Landsat (ESRI Living Atlas)

14. Paste Screenshot of summer image here (2 pts):



15. Paste screenshot of winter image here (2 pts):

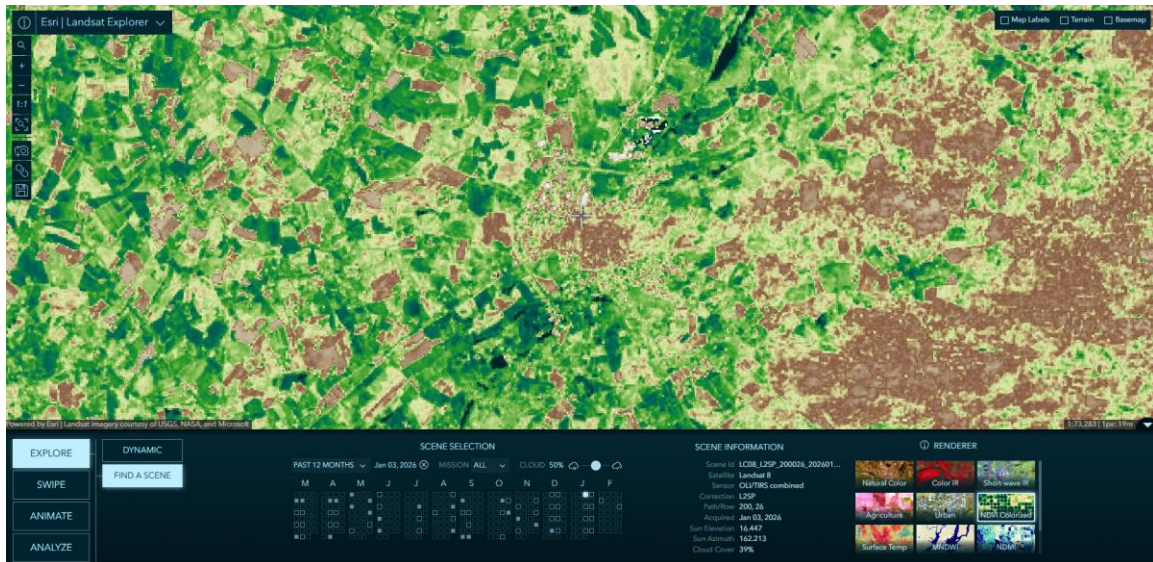


Normalized Difference Vegetation Index:

16. Paste Screenshot of summer image here (2 pts):



17. Paste screenshot of winter image here (2 pts):



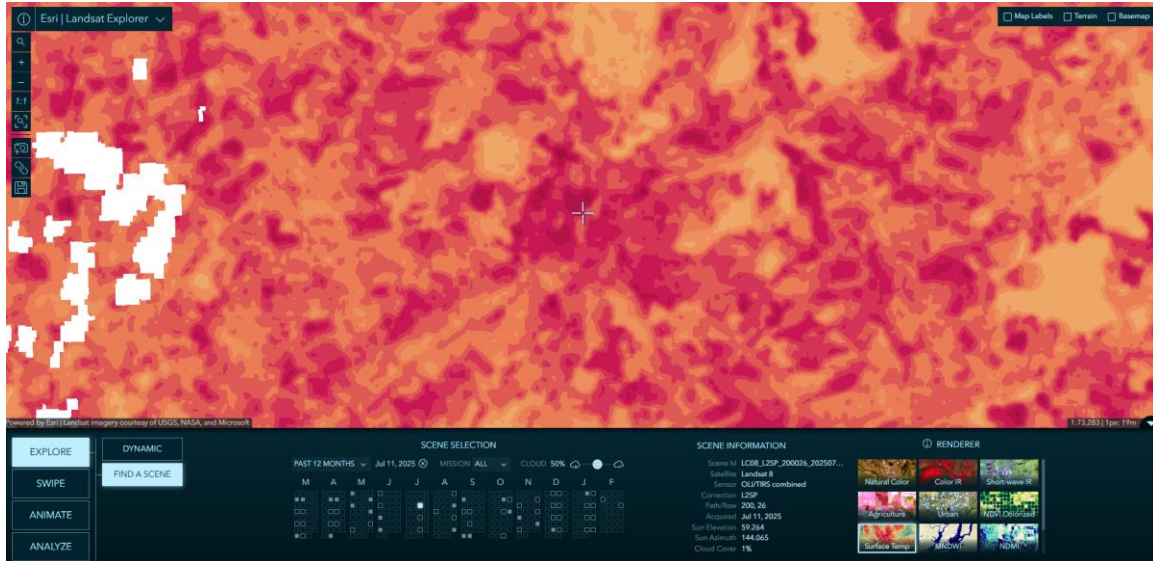
Answer:

18. How does NDVI change seasonally, does it look like the Sentinel-2 images (2 pts)?

NDVI looks like it changes seasonally the same as it does with the Sentinel-2 images, where the trees/hedgerows are greener but the fields are browner in the summer, and the opposite for the winter.

Thermal (Landsat only):

19. Paste Screenshot of summer image here (2 pts):



20. Paste screenshot of winter image here (2 pts):



Answer:

21. Which surfaces are warmest/coolest (2 pts)?

The urban spaces with buildings and roads are clearly the warmest surfaces, and it appears that the forests to the east of town are the coldest surfaces.

22. Why are these locations cooler or warmer at each time of year (2 pts)?

My guess would be the seasonal change in distance to the sun (and more importantly, the angle at which energy from the sun hits the atmosphere and then the planet surface and back), where the energy from the sun is a lot more direct in summer than it is in winter for the northern hemisphere.

Temporal resolution:

23. How often does Landsat revisit (2 pts)?

Landsat revisits this location once a week, but the interesting thing I noticed is that Landsat often takes images on back-to-back days.

24. How does this compare to Sentinel-2 (2 pts)?

Compared to Sentinel-2, it seems that Landsat revisits this area with less frequency than Sentinel-2 for the same cloud-cover parameters. At low cloud-cover levels, both Landsat and Sentinel-2 are infrequent (1-2 times per month). But with loosening the filter to allow for higher levels of cloud cover, the Landsat imagery is still only once a week compared to several times a week with Sentinel-2.

Part 4 – Comparison & Reflection

25. Which platform is better for fine-scale features (2 pts)?

The resolution to be able to see fine-scale features seemed better in the Sentinel-2 imagery. It was easier to make out the building and road features in Sentinel-2, and the image quality was overall 'sharper' due to having more pixels.

26. What does NDVI add beyond RGB (2pts)?

NDVI adds Near-Infrared (NIR) which is highly reflected by healthy vegetation. This NIR reflection enables us to better determine plant health with NDVI than with True Color imagery which only has RGB visible bands.

27. What unique info does thermal imagery provide (2 pts)?

Thermal imagery provides visibility on the heat energy emitted from surfaces (rather than measuring the reflected light from the sun), and is able to capture both day and night as well as see through smoke/haze.

28. Which platform (Sentinel-2 or Landsat) would you choose for vegetation phenology (the timing of leaf-on and leaf-off for a deciduous forest and *why* (2 pts)? Consider both spatial and temporal resolution

Personally I would choose Sentinel-2, especially if I am studying an area in Europe/Northern Hemisphere. The spatial resolution is slightly better to provide finer details about the vegetation. And the temporal resolution provides slightly more frequent imagery of the same location which lends itself better to analysis over time (and in this case specifically, we need as much frequency as we can get to see as near as possible to the date that trees in a forest shed/gained leaves.